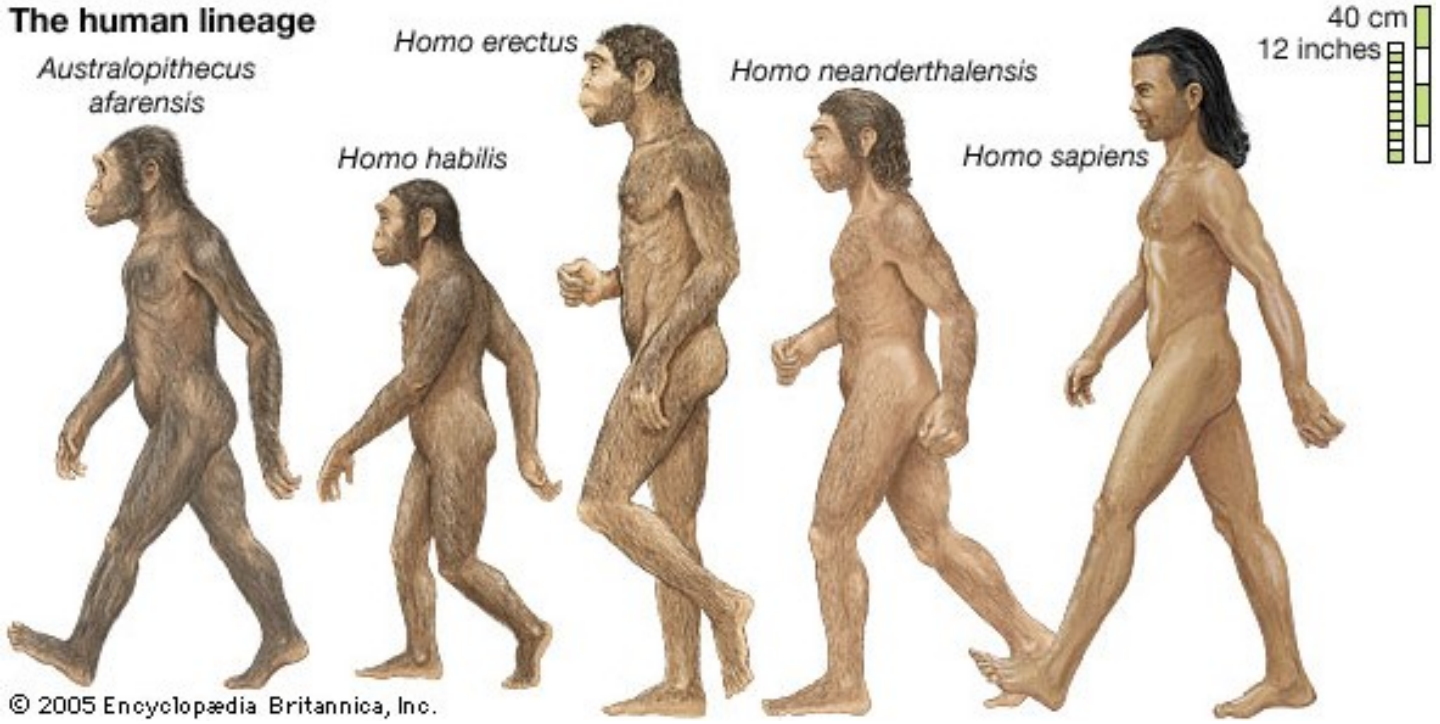
Hominin Evolution



Intro

Adaptations for Erect Posture

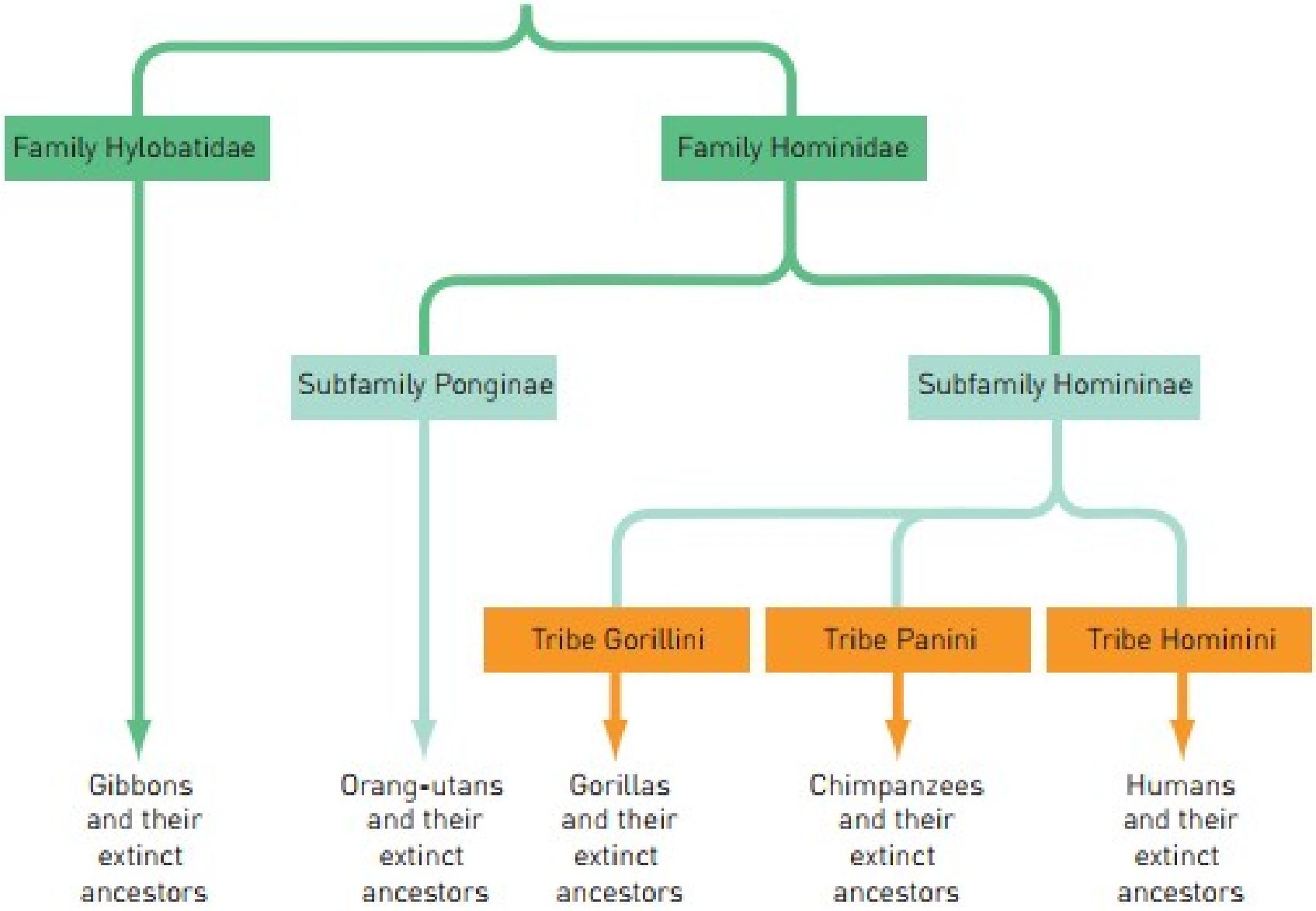


Figure 18.2 Classification of apes and humans

Bipedalism – walking on two legs

- Humans have unique locomotion compared to other primate species.
  - Other primates (eg monkeys, gorillas, chimps) are *quadrupedal*.
  - Humans have evolved to be *bipedal*.
- Some extinct human ancestors must have been bipedal and this **must have been an advantage in the environment.**
- First evidence of ancestral bipedalism – The Laetoli Footprints – 3 million years old
  - Made by *Australopithecus* – footprints fossilised in volcanic ash
  - Only 2 feet in prints
  - Feet showed adaptations for bipedalism (more on this later)
  - *Note – this does not mean that Australopithecus was the first bipedal human ancestor – just the first we have excellent evidence for. Earlier ancestors would have shown a slow sequence of changes towards bipedalism*



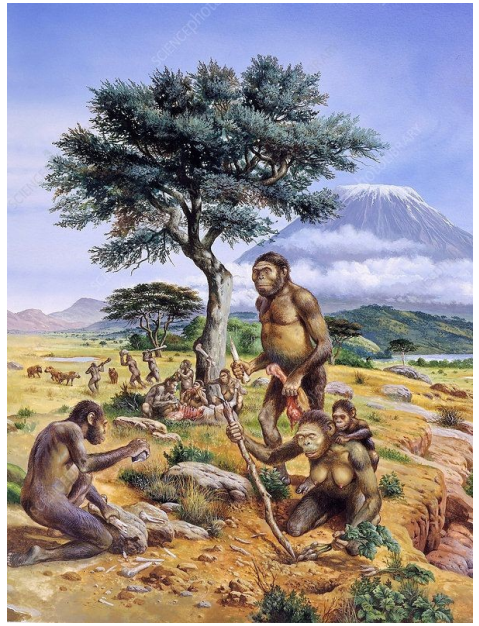


Learning Aim: Define bipedalism and describe evidence that recent human ancestors were bipedal.

Selection Pressures and Evolution of Bipedalism

- Evolution by natural selection means environment must have favoured bipedalism for human ancestors:
  - Hominins that evolved into human ancestors originally lived in woodland environment
    - Ape-like
    - Arms and hands suited to arboreal lifestyle
    - May have used some bipedal movement to get through forest clearings – like modern Orangutans.
  - Environmental change in Africa 5-6 million years ago
    - Temperatures fell and rainfall decreased
    - Forest areas diminished
    - Open grassland increased
    - Slow evolutionary changes towards bipedalism would have been favoured.





Learning Aim: Discuss environmental changes that favoured bipedalism.

Selection Pressures and Evolution of Bipedalism

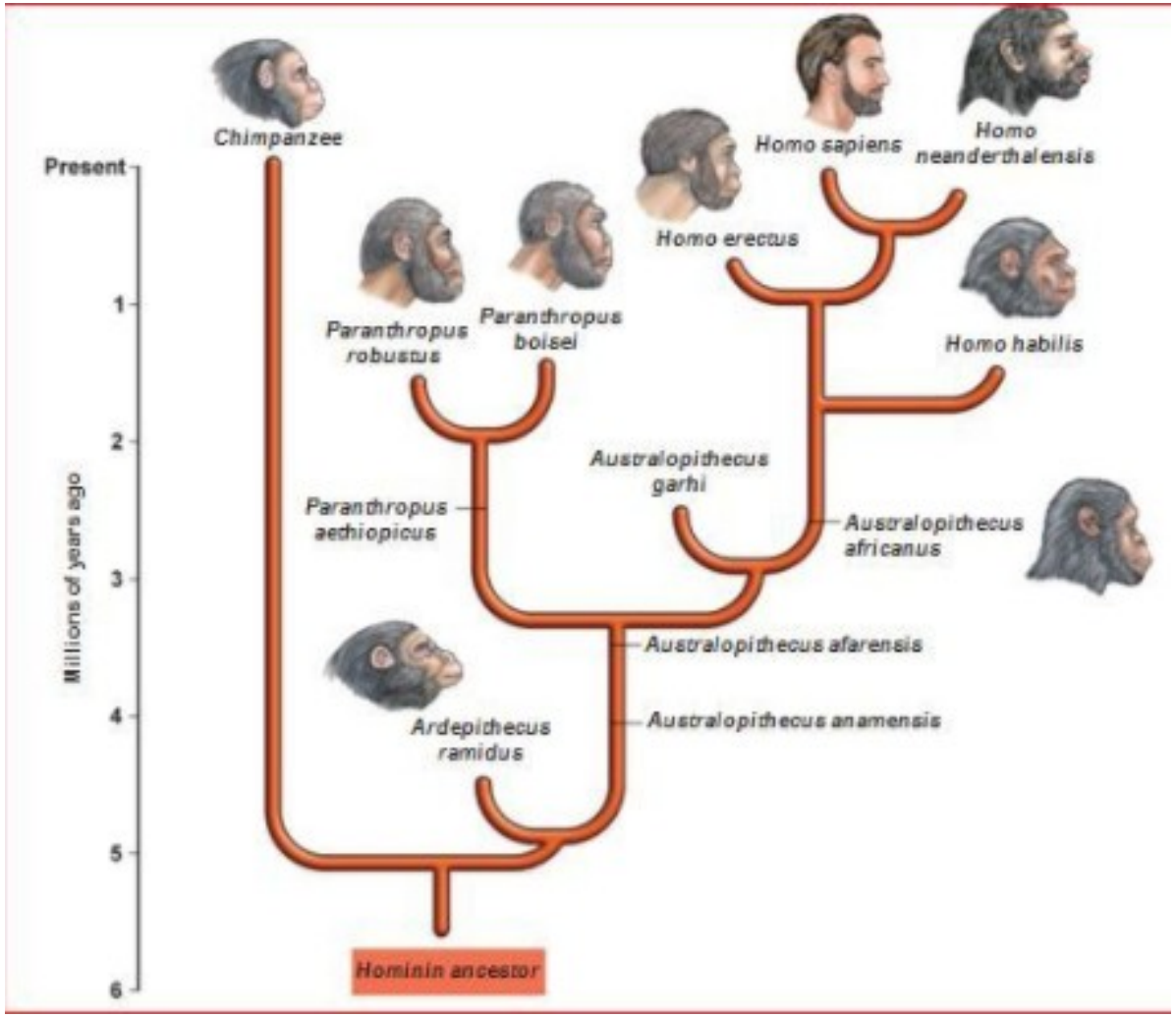
- Increase in open grassland caused bipedalism and other traits to be advantageous
  - Bipedalism is a more energy efficient mode of locomotion on the ground
  - Increased range of vision to detect prey and predators
  - Increased height/size deterred predators
  - Hands free for carrying food, using tools etc
  - Higher reach to pick fruit or collect honey
  - Less exposure to sunlight, greater surface areas for breezes – better temperature regulation in open grassland.

- Food supply limited as forests replaced by grasslands
  - Caused selection pressure and made bipedalism a big selective advantage

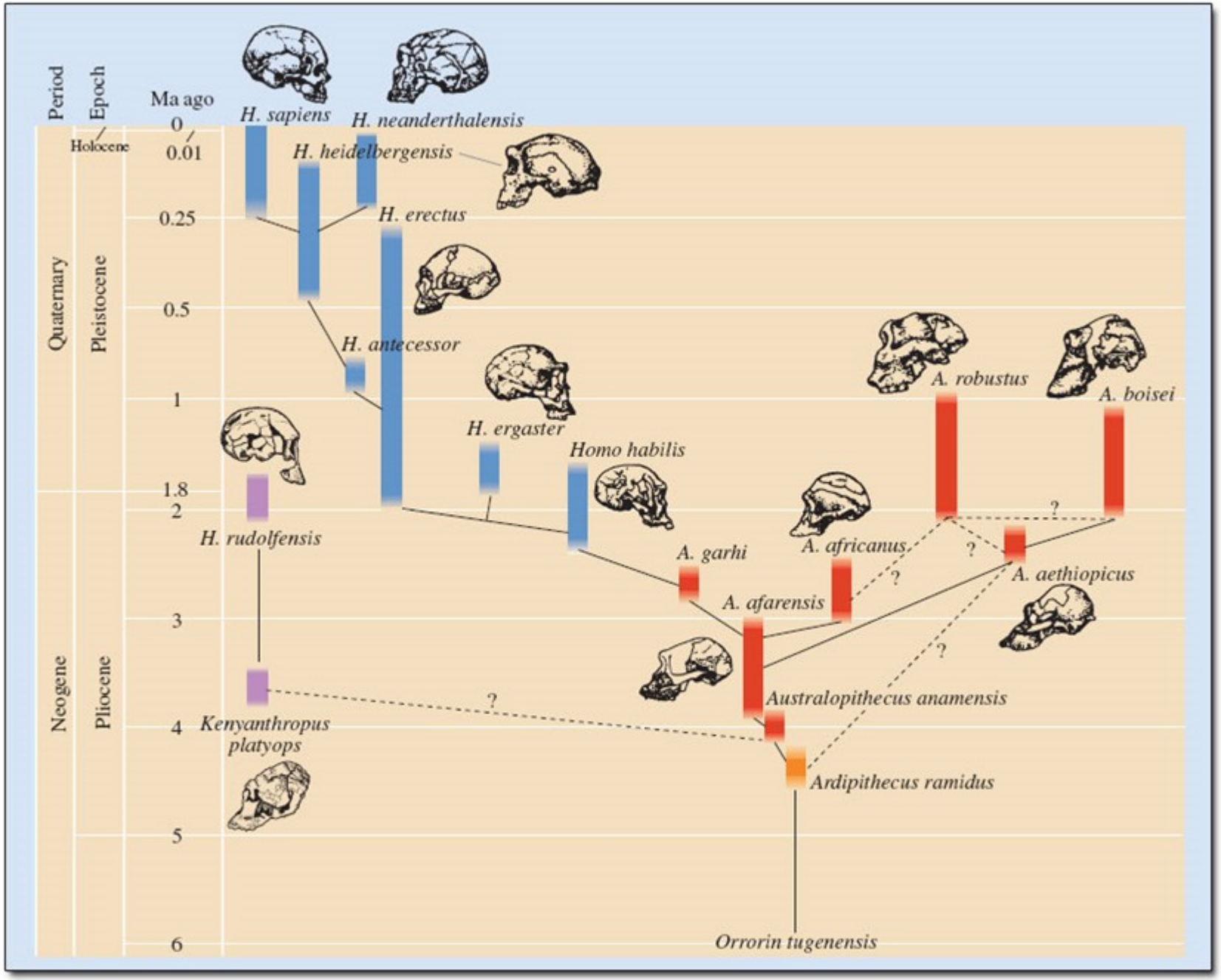
Learning Aim: Explain why bipedalism was an advantage in an open grassland environment.

Bipedal Human Ancestors

- Bipedal human ancestors
  - Extinct ancestors of present-day humans
  - Apes alive today and humans share a common ancestor. Human lineages and ape lineages diverged from that common ancestor.
  - Hominids of interest to this course:
    - *Australopithecus*
    - *Homo habilis*
    - *Homo erectus*
    - *Homo neanderthalensis*
    - *Homo sapiens*



Learning Aim: Review extinct bipedal human ancestors and when they existed.


Period
Epoch
Ma ago
Quaternary
Pleistocene
Holocene
Neogene
Pliocene
0
0.01
0.25
0.5
1
1.8
2
3
4
5
6
H. sapiens
H. neanderthalensis
H. heidelbergensis
H. erectus
H. antecessor
H. ergaster
Homo habilis
H. rudolfensis
A. garhi
A. africanus
A. robustus
A. boisei
A. aethiopicus
A. afarensis
Australopithecus anamensis
Kenyanthropus platyops
Ardipithecus ramidus
Orrorin tugenensis
?

Adaptations for Bipedalism/Erect Posture

- Skeleton and muscles of humans are adapted for erect posture to enable bipedal locomotion
- Extinct bipedal human ancestors show a progression in these adaptations
- Can see these adaptations by comparing humans with our close quadrupedal relative - the gorilla.
- Adaptations for erect posture/bipedalism include:

  - Position of Foramen Magnum and Skull
  - Curve of spinal Column
  - Jaw (reduced prognathism)
  - Pelvis
  - Carrying angle
  - Knee
  - Foot
  - Centre of Gravity
  - Stance and muscle tone
  - Striding gait

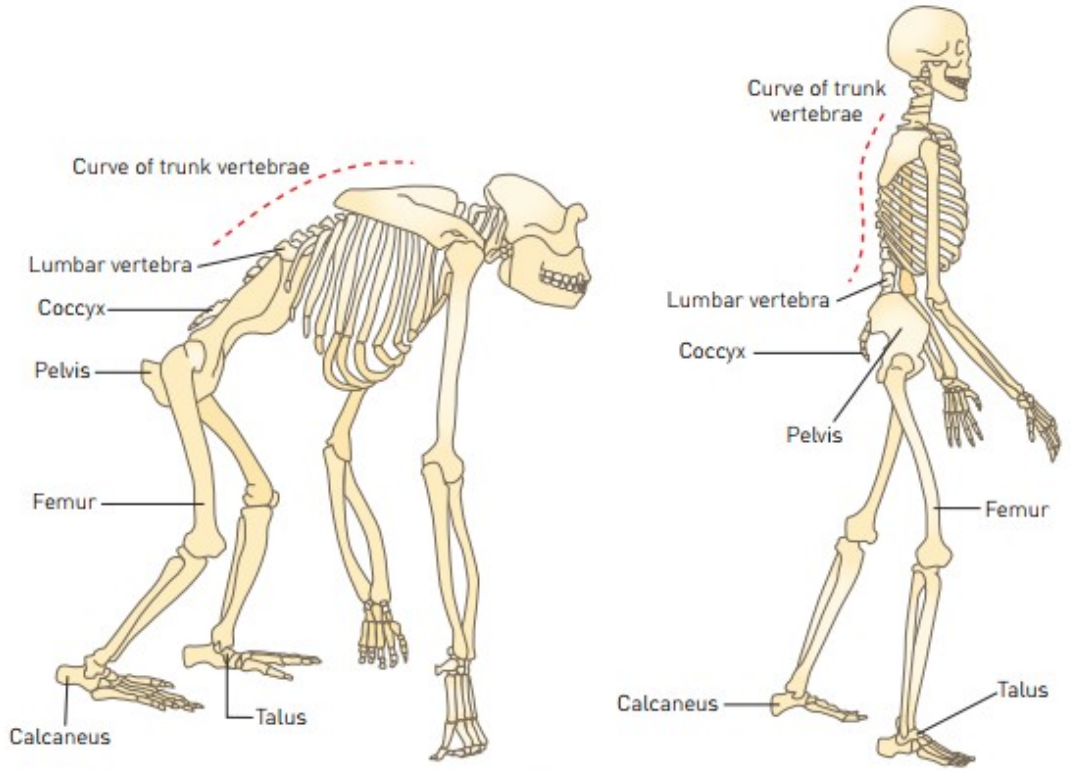


Figure 18.3 Posture of the skeletons of a gorilla and a human

Learning Aim: List anatomical adaptations for bipedalism/erect posture.

Adaptations for Bipedalism – position of Foramen Magnum

- Foramen Magnum
  - Hole in base of skull where spinal cord joins brain
  - Quadrupeds: Foramen magnum is towards back of skull
  - In Bipeds the Foramen Magnum sits at the centre of the base of the skull
    - Skull sits atop spinal column – balances.
    - Position of skull reduces force and muscle needed to hold head up.
    - Can see evolutionary progression in extinct ancestors

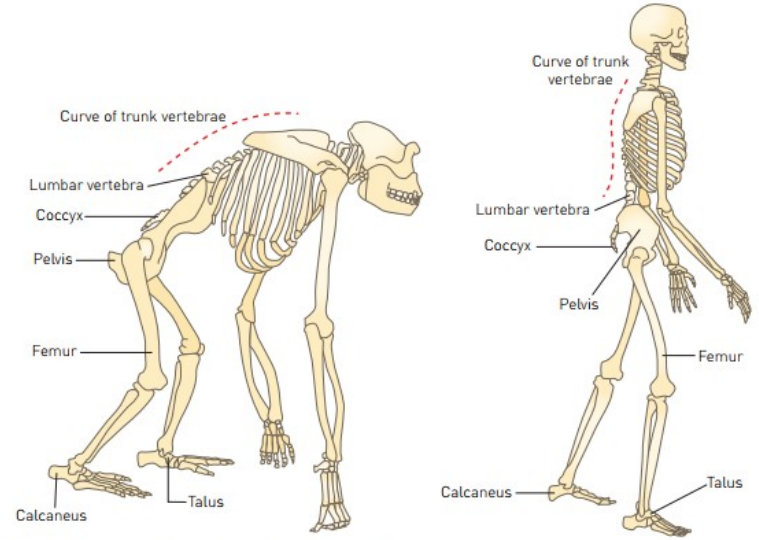
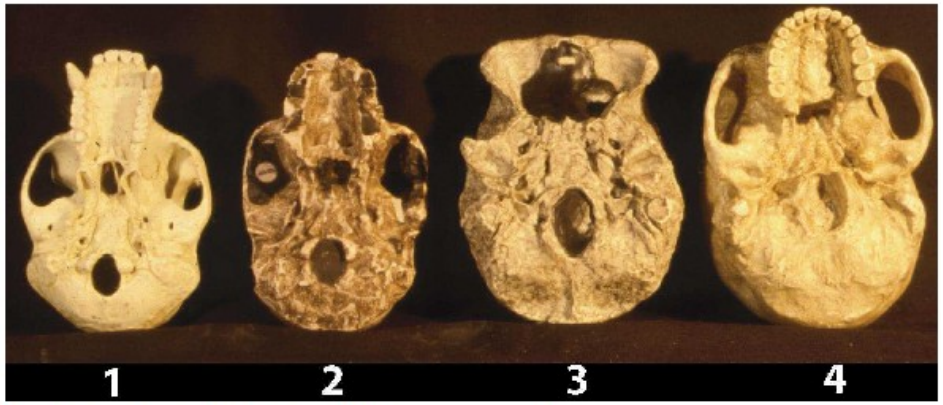
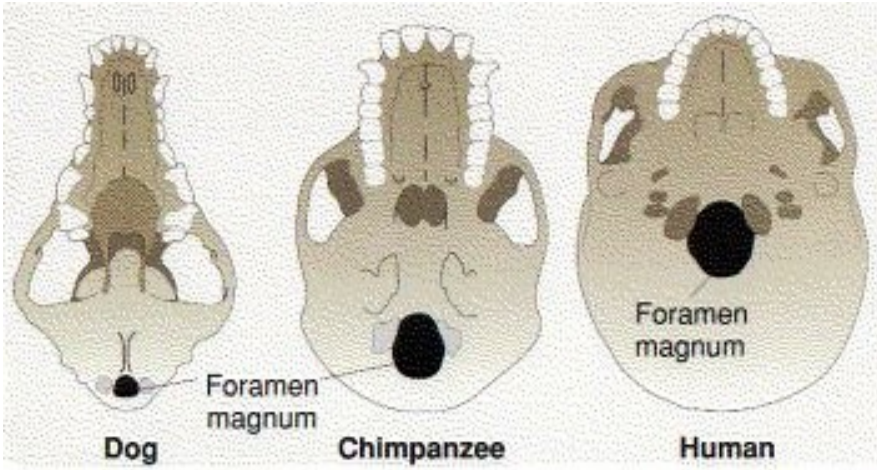


Figure 18.3 Posture of the skeletons of a gorilla and a human



1: *Pan troglodytes* (Chimpanzee)
2: *Australopithecus*
3: *Homo erectus*
4: *Homo sapiens* - (human)

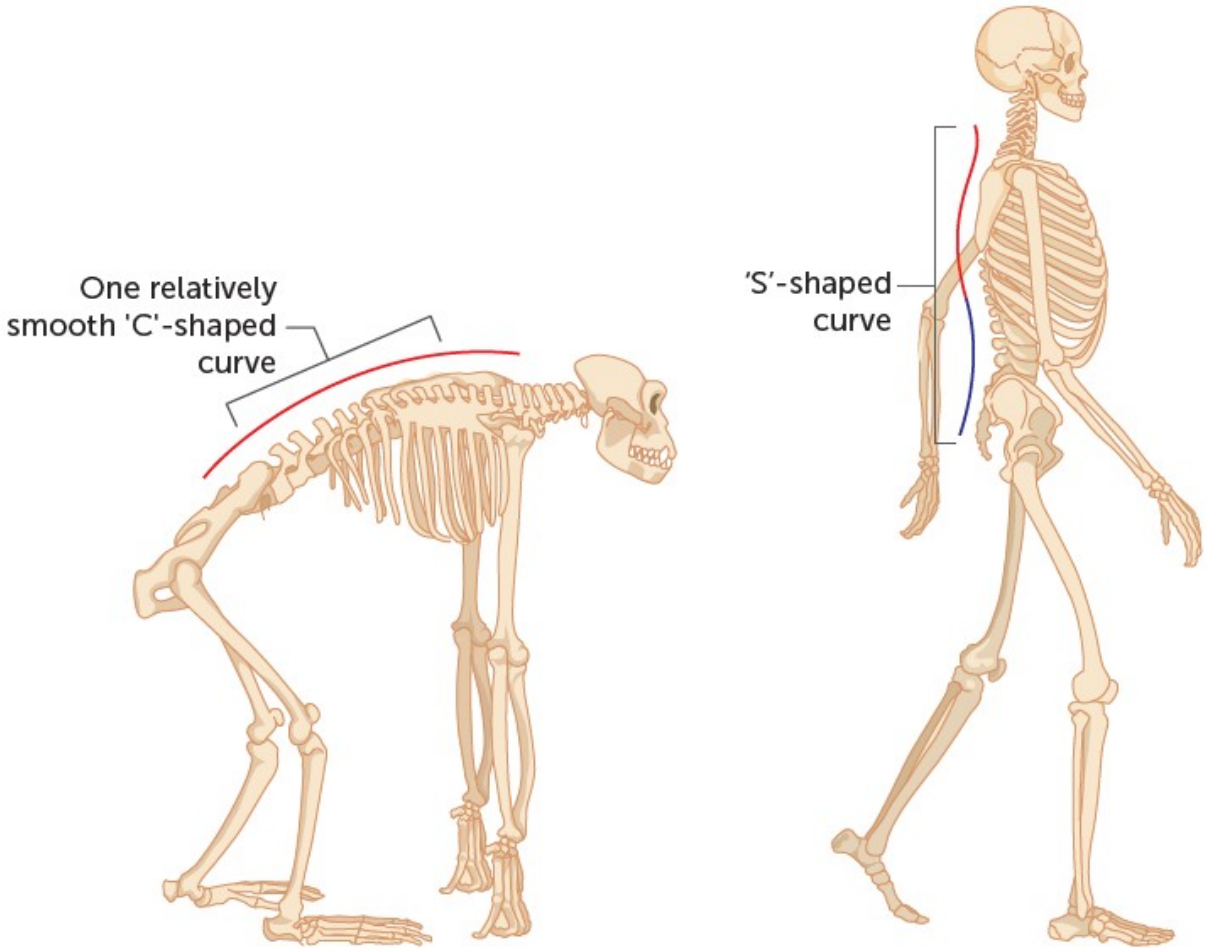


Learning Aim: Describe how the foramen magnum is positioned for bipedalism and why this is an advantage.

Adaptations for Bipedalism– Curvature of Spinal Column

- Humans: double curvature – S shape
  - Contributes to upright stance
  - Lumbar spine: vertebrae wedge shaped – forward curve
  - Improves body balance while upright
  - Cervical curve brings centre of gravity of skull over spine.

- Compare to quadruped apes – C shaped spine

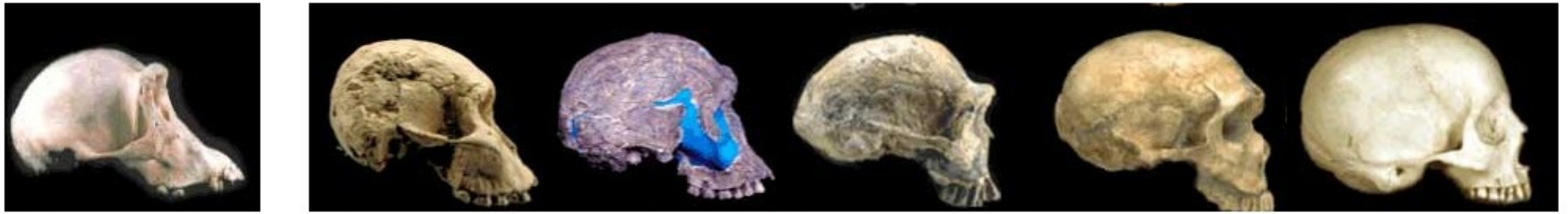


Learning Aim: Describe how spinal curvature is adapted for bipedalism and why this is an advantage.

Adaptations for Bipedalism – reduced prognathism

- Prognathism = protruding jaw
  - Apes are prognathic – jaw juts out.
  - Humans: jaw is much flatter to face
    - Allows skull to balance better on spinal cord – prevents front of skull from being heavier than back – less muscle needed to raise head.
    - Jaw size also significant for other reasons – more on this later





Learning Aim: Describe adaptations to the jaw for bipedalism and why this is an advantage.

Adaptations for Bipedalism – The Pelvis

- Humans: have a broader, flatter, shorter, bowl shaped pelvis than quadrupedal apes. This pelvis can be seen in bipedal human ancestors, trending more bowl shaped over time.

- Broad, flat, short, bowl shaped pelvis:
  - Supports abdominal organs when standing up
  - Supports developing foetus in female
  - Female pelvis slightly broader than male – birth
  - Broad hip bones – attachment for large buttocks – move legs, keeps upper body erect

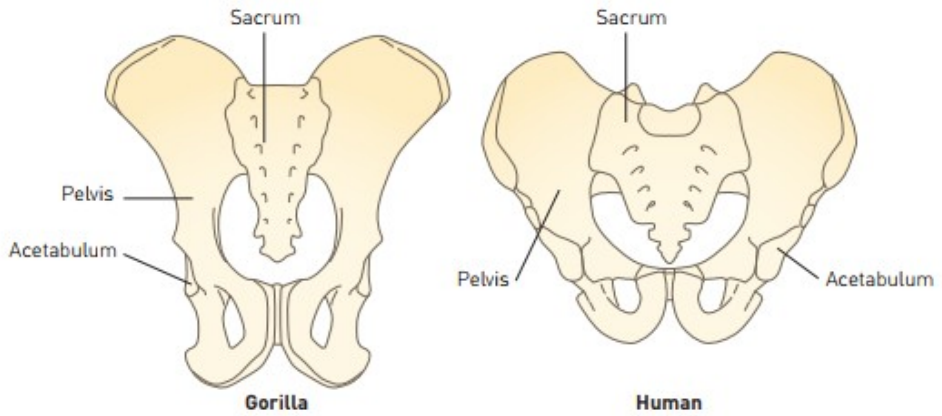
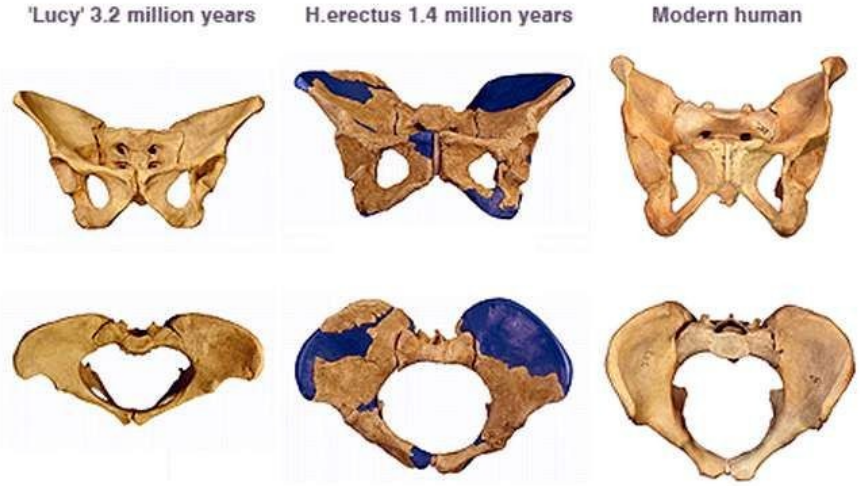


Figure 18.7 Pelves of a gorilla and a human



Learning Aim: Describe how the pelvis is adapted for bipedalism and why this is an advantage.

Adaptations for Bipedalism – Carrying Angle

- The Carrying Angle
  - Refers to the positioning of bones and muscles in legs, to carry body weight.

  In humans the carrying angle is adapted for bipedalism
  - hip joint directly under trunk and head
    - Large femoral head
    - Fits into well formed hip socket (acetabulum)
    - Hip sockets wide apart
  - Femurs converge (are closer together) towards knees
    - At angle to vertical
    - This is called the “carrying angle”
    - Ensures weight distribution stays close to central axis of body
    - Allows upright, stable, striding gait, rather than side to side rocking gait seen in apes
    - Weight transmission falls outside femur

Compare with bipedal chimps – see diagrams

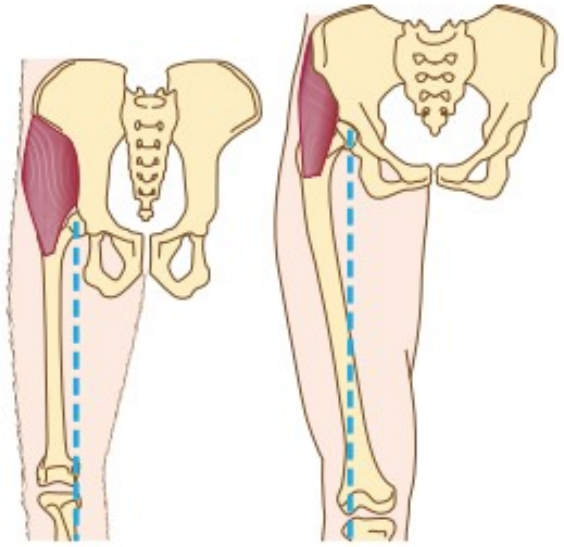
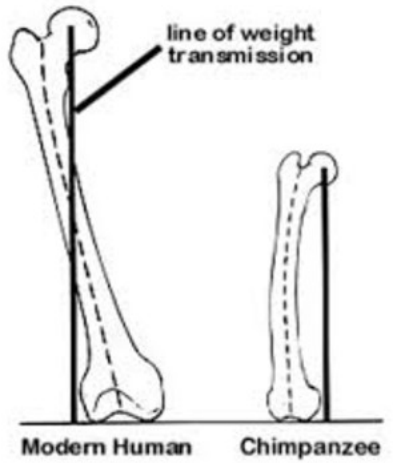


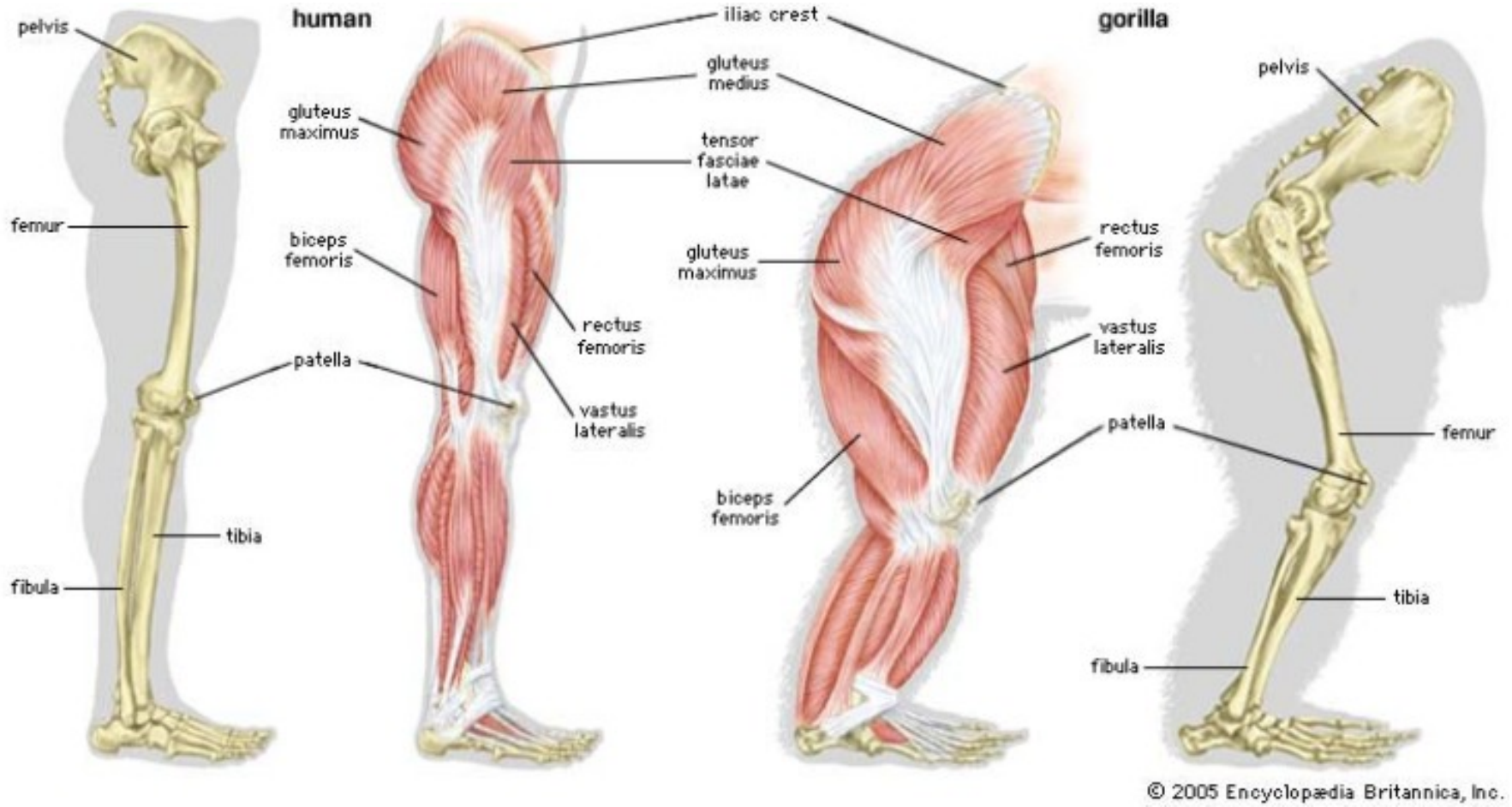
Figure 18.8 The pelves and femurs of chimpanzees and humans, showing how humans have a carrying angle, with the femur angled in towards the knee (the dotted line shows the direction of weight transmission)



Learning Aim: Describe how the carrying angle of the femur is adapted for bipedalism and why this is an advantage.

Adaptations for Bipedalism – Carrying Angle

Note that in humans, the hip joint is in line with the abdomen, chest and head, allowing for upright posture.



Adaptations for Bipedalism – the Knee

- In bipedal humans and human ancestors:
  - Weight transmitted down outside of femur to knee
  - 2 part hinge joint angled for weight transmission. Shaping of joint allows outer aspect to be stronger.
  - Centre of gravity falls just in front of knees
    - Force pushes to bend knee backwards, resisted by knee ligaments
    - Natural resistance of joint requires no energy to hold body in standing position
  - Compare to quadrupedal hominins (see picture)



Homo sapiens *Australopithecus* Chimpanzee

Learning Aim: Describe how the knee is adapted for bipedalism and why this is an advantage.

Adaptations for Bipedalism – the foot (and lower leg)

- In bipedal hominins, from the knee joint, the weight is transmitted through the tibia to the ankle.
- Tibia larger and stronger than fibula.
- At ankle, body weight transmitted from tibia through ankle bone (talus) to other tarsal bones, to metatarsals and phalanges, via foot arch.
- Bipedal foot distinctively adapted
  - "Platform foot"
  - Has lost flexibility and therefore prehensility
  - Large big toe, not opposable, aligned with other toes
  - Metatarsals form arches – longitudinal and transverse
  - Allows striding gait as foot "rolls" with movement.

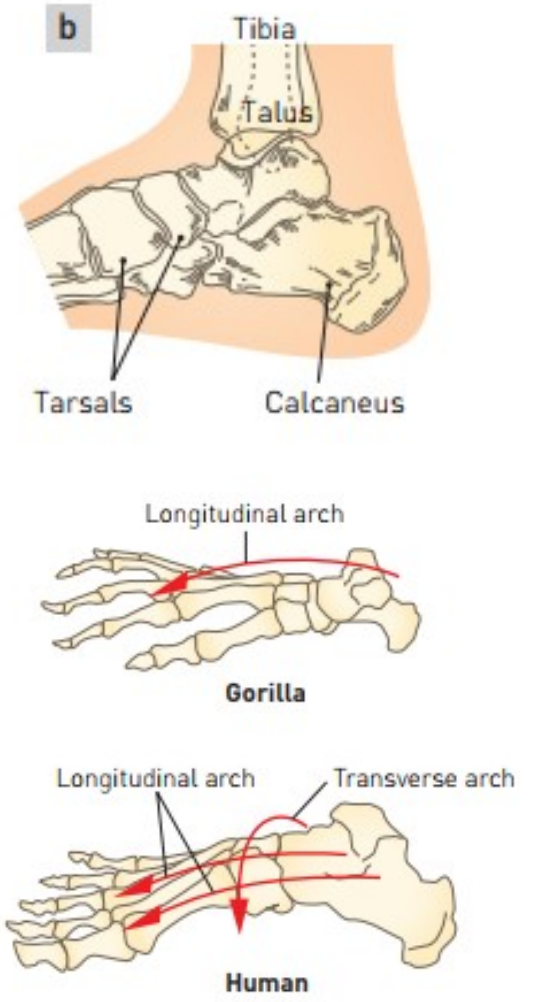


Figure 18.10 The arches of the foot of a gorilla and a human: humans have a longitudinal arch and a transverse arch; gorillas only the former.

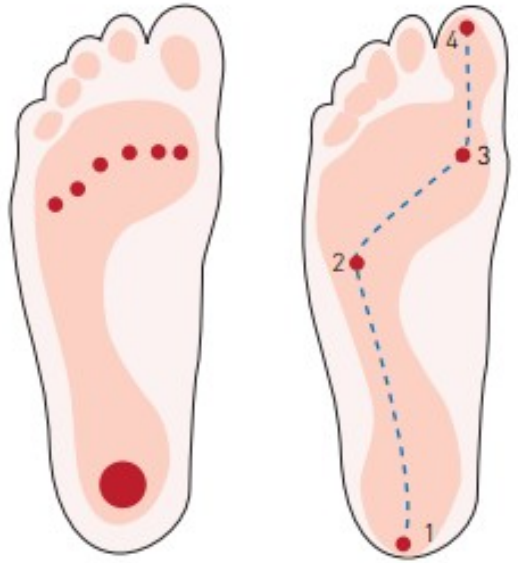


Figure 18.13 The diagram on the left shows how body weight is borne by the foot when standing still. The one on the right shows the distribution of body weight as a step is taken: the weight of the body is progressively borne on points 1 to 4 as the heel of the foot hits the ground and the big toe thrusts off.

Learning Aim: Describe how the foot is adapted for bipedalism and why this is an advantage.

Adaptations for Bipedalism – Centre of Gravity

- Low Centre of Gravity makes balancing easier.
- Humans: legs longer than arms.
  - (1/2 of height vs 1/3 of height in chimps)
  - Long legs increase stride length
  - Lowers the centre of gravity to pelvis (chest in chimps)
  - Lower centre of gravity contributes to stability when standing/walking

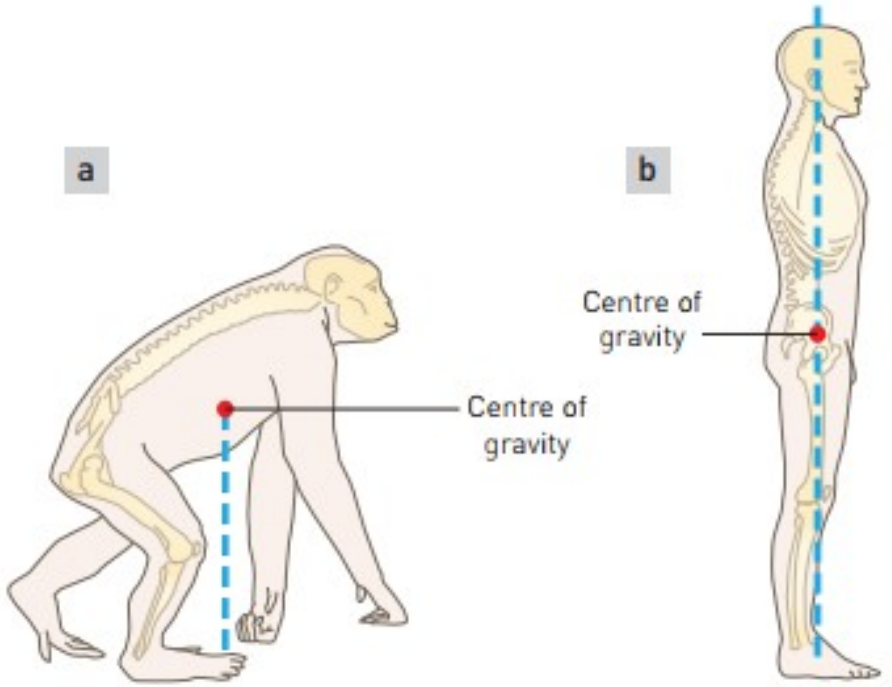


Figure 18.11 Centre of gravity of **a** an ape and **b** a human

Learning Aim: Describe how the centre of gravity is adapted for bipedalism and why this is an advantage.

Adaptations for Bipedalism - Stance and Muscle Tone

- Muscle Tone
  - Essential for maintaining an upright stance (erect posture).
  - Consistent and partial contraction of skeletal muscles.
  - Muscles required for balance in upright stance need greater muscle tone:
    - Spinal muscles
    - Hip, knee, ankle
    - Abdominals
  - Nervous system and sense organs work together to maintain muscle tone

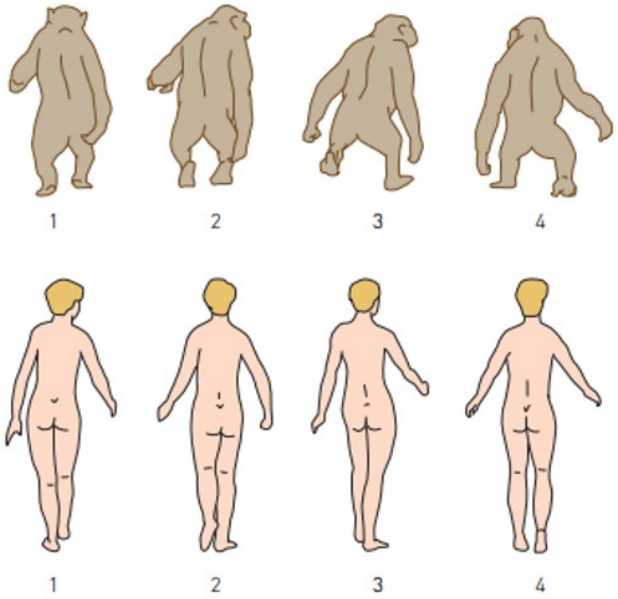


Figure 18.12 Muscles supporting the body against gravity

Learning Aim: Describe how stance and muscle tone are adapted for bipedalism and why this is an advantage.

Adaptations for Bipedalism – Striding Gait

- Walking upright with hip and knee fully straightened.
- Only bipedal hominins (humans and extinct recent ancestors) have this form of locomotion.
- Other hominids bend forward at hips, knees bent, swaying gait.
- Arches of bipedal foot allow rolling movement, heel to big toe
- Human big toe no longer prehensile or opposable: allows stability for weight-bearing in striding gait.
- Trunk rotates around pelvis
  - Arms swing forwards
  - Shoulders at right angles to direction of movement
- Arrangement of femurs allows weight distribution close to central axis



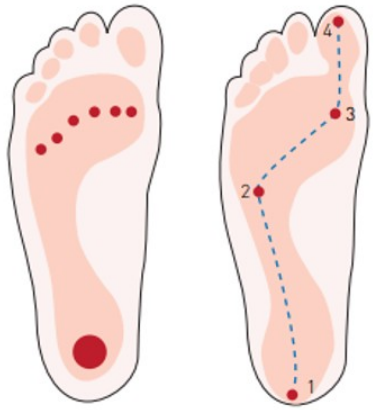


Figure 18.13 The diagram on the left shows how body weight is borne by the foot when standing still. The one on the right shows the distribution of body weight as a step is taken: the weight of the body is progressively borne on points 1 to 4 as the heel of the foot hits the ground and the big toe thrusts off.

Learning Aim: Describe how gait is adapted for bipedalism and why this is an advantage.